

Liniaizarea modelelor neliniare

Majoritatea proceselor chimice sunt reprezentate de modele matematice neliniare

Se consideră un model neliniar format dintr-o singură variabilă:

$$\frac{dX}{dt} = f(X)$$

Funcția $f(x)$ poate fi aproximată folosind descompunerea în serie Taylor în jurul punctului static de funcționare (x_s), cu neglijarea termenilor de grad superior:

$$f(X) = f(X_s) + \left. \frac{\partial f}{\partial X} \right|_{X_s} (X - X_s) + \frac{1}{2} \left. \frac{\partial^2 f}{\partial X^2} \right|_{X_s} (X - X_s)^2 + \text{termenii de grad superior}$$

După eliminarea termenilor de grad superior, forma funcției $f(x)$ este:

$$f(X) \approx f(X_s) + \left. \frac{\partial f}{\partial X} \right|_{X_s} (X - X_s)$$

derivate parțială a funcției f în raport cu x
evaluate în punctul x_s

În regim staționar:

$$\frac{dX_s}{dt} = f(X_s) = 0$$

$$\frac{dX}{dt} = f(X) \approx \left. \frac{\partial f}{\partial X} \right|_{X_s} (X - X_s)$$

Deoarece derivata unei constante este zero:

$$\frac{dX}{dt} = \frac{d(X - X_s)}{dt}$$

$$\frac{d(X - X_s)}{dt} \approx \left. \frac{\partial f}{\partial X} \right|_{X_s} (X - X_s)$$

În final se obține următoarea formă:

$$\frac{d\bar{X}}{dt} \approx \left. \frac{\partial f}{\partial X} \right|_{X_s} \bar{X}$$

$$\bar{X} = X - X_s$$

$$\frac{d\bar{X}}{dt} \approx a \bar{X} \quad \text{where} \quad a = \left. \frac{\partial f}{\partial X} \right|_{X_s}$$

În continuare se consideră o funcție formată dintr-o variabilă de stare și o mărime de intrare.

$$\dot{X} = \frac{dX}{dt} = f(X, u)$$

$$\begin{aligned} \dot{X} = f(X_s, u_s) &+ \left. \frac{\partial f}{\partial X} \right|_{X_s, u_s} (X - X_s) + \left. \frac{\partial f}{\partial u} \right|_{X_s, u_s} (u - u_s) + \frac{1}{2} \left. \frac{\partial^2 f}{\partial X^2} \right|_{X_s, u_s} (X - X_s)^2 \\ &+ \left. \frac{\partial^2 f}{\partial X \partial u} \right|_{X_s, u_s} (X - X_s)(u - u_s) + \frac{1}{2} \left. \frac{\partial^2 f}{\partial u^2} \right|_{X_s, u_s} (u - u_s)^2 \cdot \quad \begin{array}{l} \text{+ termenii} \\ \text{de grad} \\ \text{superior} \end{array} \end{aligned}$$

$$\dot{x} \approx \underbrace{f(x_s, u_s)}_{\text{zero}} + \left. \frac{\partial f}{\partial x} \right|_{x_s, u_s} (x - x_s) + \left. \frac{\partial f}{\partial u} \right|_{x_s, u_s} (u - u_s)$$

$$\frac{d(x - x_s)}{dt} \approx \left. \frac{\partial f}{\partial x} \right|_{x_s, u_s} (x - x_s) + \left. \frac{\partial f}{\partial u} \right|_{x_s, u_s} (u - u_s)$$

$$\frac{d\bar{x}}{dt} \approx \left. \frac{\partial f}{\partial x} \right|_{x_s, u_s} \bar{x} + \left. \frac{\partial f}{\partial u} \right|_{x_s, u_s} \bar{u}$$

$$\frac{d\bar{x}}{dt} \approx a\bar{x} + b\bar{u} \quad a = \left. \frac{\partial f}{\partial x} \right|_{x_s, u_s} \quad b = \left. \frac{\partial f}{\partial u} \right|_{x_s, u_s}$$

Unde

$$y = g(x, u)$$

$$g(x, u) \approx g(x_s, u_s) + \left. \frac{\partial g}{\partial x} \right|_{x_s, u_s} (x - x_s) + \left. \frac{\partial g}{\partial u} \right|_{x_s, u_s} (u - u_s) \quad g(x_s, u_s) = y_s$$

$$y - y_s = \left. \frac{\partial g}{\partial x} \right|_{x_s, u_s} (x - x_s) + \left. \frac{\partial g}{\partial u} \right|_{x_s, u_s} (u - u_s)$$

$$\bar{y} = c\bar{x} + d\bar{u} \quad \text{where} \quad c = \left. \frac{\partial g}{\partial x} \right|_{x_s, u_s} \quad d = \left. \frac{\partial g}{\partial u} \right|_{x_s, u_s}$$

Liniaizarea sistemelor cu mai multe variabile de stare:

$$\dot{x}_1 = \frac{dx_1}{dt} = f_1(x_1, x_2, u)$$

$$y = g(x_1, x_2, u)$$

$$\dot{x}_2 = \frac{dx_2}{dt} = f_2(x_1, x_2, u)$$

$$f_1(x_1, x_2, u) = f_1(x_{1s}, x_{2s}, u_s) + \left. \frac{\partial f_1}{\partial x_1} \right|_{x_{1s}, x_{2s}, u_s} (x_1 - x_{1s}) + \left. \frac{\partial f_1}{\partial x_2} \right|_{x_{1s}, x_{2s}, u_s} (x_2 - x_{2s}) + \left. \frac{\partial f_1}{\partial u} \right|_{x_{1s}, x_{2s}, u_s} (u - u_s)$$

$$f_2(x_1, x_2, u) = f_2(x_{1s}, x_{2s}, u_s) + \left. \frac{\partial f_2}{\partial x_1} \right|_{x_{1s}, x_{2s}, u_s} (x_1 - x_{1s}) + \left. \frac{\partial f_2}{\partial x_2} \right|_{x_{1s}, x_{2s}, u_s} (x_2 - x_{2s}) + \left. \frac{\partial f_2}{\partial u} \right|_{x_{1s}, x_{2s}, u_s} (u - u_s)$$

$$g(x_1, x_2, u) = g(x_{1s}, x_{2s}, u_s) + \left. \frac{\partial g}{\partial x_1} \right|_{x_{1s}, x_{2s}, u_s} (x_1 - x_{1s}) + \left. \frac{\partial g}{\partial x_2} \right|_{x_{1s}, x_{2s}, u_s} (x_2 - x_{2s}) + \left. \frac{\partial g}{\partial u} \right|_{x_{1s}, x_{2s}, u_s} (u - u_s)$$

$$f_1(x_{1s}, x_{2s}, u_s) = f_2(x_{1s}, x_{2s}, u_s) = 0 \quad g(x_{1s}, x_{2s}, u_s) = y_s$$

$$\frac{dx_1}{dt} = \frac{d(x_1 - x_{1s})}{dt} \quad \frac{dx_2}{dt} = \frac{d(x_2 - x_{2s})}{dt}$$

$$\begin{bmatrix} \frac{d(x_1 - x_{1s})}{dt} \\ \frac{d(x_2 - x_{2s})}{dt} \end{bmatrix} = \begin{bmatrix} \left. \frac{\partial f_1}{\partial x_1} \right|_{x_{1s}, x_{2s}, u_s} & \left. \frac{\partial f_1}{\partial x_2} \right|_{x_{1s}, x_{2s}, u_s} \\ \left. \frac{\partial f_2}{\partial x_1} \right|_{x_{1s}, x_{2s}, u_s} & \left. \frac{\partial f_2}{\partial x_2} \right|_{x_{1s}, x_{2s}, u_s} \end{bmatrix} \begin{bmatrix} x_1 - x_{1s} \\ x_2 - x_{2s} \end{bmatrix} + \begin{bmatrix} \left. \frac{\partial f_1}{\partial u} \right|_{x_{1s}, x_{2s}, u_s} \\ \left. \frac{\partial f_2}{\partial u} \right|_{x_{1s}, x_{2s}, u_s} \end{bmatrix} [u - u_s]$$

$$y - y_s = \begin{bmatrix} \left. \frac{\partial g}{\partial x_1} \right|_{x_{1s}, x_{2s}, u_s} & \left. \frac{\partial g}{\partial x_2} \right|_{x_{1s}, x_{2s}, u_s} \end{bmatrix} \begin{bmatrix} x_1 - x_{1s} \\ x_2 - x_{2s} \end{bmatrix} + \left. \frac{\partial g}{\partial u} \right|_{x_{1s}, x_{2s}, u_s} [u - u_s]$$

$$\dot{\bar{\mathbf{x}}} = \mathbf{A} \bar{\mathbf{x}} + \mathbf{B} \bar{\mathbf{u}}$$

$$\bar{\mathbf{y}} = \mathbf{C} \bar{\mathbf{x}} + \mathbf{D} \bar{\mathbf{u}}$$

Jacobian

Generalizare:

Se consider un sistem nelinier cu n variabile de stare, m variabile de intrare și r variabile de ieșire:

$$\dot{x}_1 = f_1(x_1, \cdots, x_n, u_1, \cdots, u_m)$$

$$\vdots$$

$$\dot{x}_n = f_n(x_1, \cdots, x_n, u_1, \cdots, u_m)$$

$$y_1 = g_1(x_1, \cdots, x_n, u_1, \cdots, u_m)$$

$$\vdots$$

$$y_r = g_r(x_1, \cdots, x_n, u_1, \cdots, u_m)$$

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$$

$$\mathbf{y} = \mathbf{g}(\mathbf{x}, \mathbf{u})$$

$$A_{ij} = \left. \frac{\partial f_i}{\partial x_j} \right|_{\mathbf{x}_s, \mathbf{u}_s} \qquad B_{ij} = \left. \frac{\partial f_i}{\partial u_j} \right|_{\mathbf{x}_s, \mathbf{u}_s}$$

$$C_{ij} = \left. \frac{\partial g_i}{\partial x_j} \right|_{\mathbf{x}_s, \mathbf{u}_s} \qquad D_{ij} = \left. \frac{\partial g_i}{\partial u_j} \right|_{\mathbf{x}_s, \mathbf{u}_s}$$

$$\dot{\bar{\mathbf{x}}} = \mathbf{A} \bar{\mathbf{x}} + \mathbf{B} \bar{\mathbf{u}}$$

$$\bar{\mathbf{y}} = \mathbf{C} \bar{\mathbf{x}} + \mathbf{D} \bar{\mathbf{u}}$$

sau

$$\dot{\mathbf{x}} = \mathbf{A} \mathbf{x} + \mathbf{B} \mathbf{u}$$

$$\mathbf{y} = \mathbf{C} \mathbf{x} + \mathbf{D} \mathbf{u}$$